02-510/710: Computational Genomics, Spring 2026

HW2: Clustering, Dimensionality Reduction, Motif Finding

Version: 1
Due: 23:59 EST, Feb 17, 2026 on Canvas

Topics in this assignment:

1. Dimensionality Reduction
2. Clustering
3. Motif Finding

What to hand in.

- One write-up (in PDF format) addressing each of the questions.
- All source code used to generate the results (scripts and/or notebooks).

For the **clustering** and **classification** problems, no skeleton code is provided and these problems will **not** be graded by an autograder. Instead, you are required to submit *working code files* (e.g., scripts or notebooks) that can produce results for these two tasks. Partial credit will be awarded based on the correctness of the methodology, implementation effort, and clarity of the results, even if the final answers are not fully correct.
Submit the following files as a single `.zip` file to Canvas:

./S2026HW2.pdf
./clustering.*

Please note that all the solutions and code must be your own. We will check for plagarism after the final submission.

1. **[40 points] Dimensionality Reduction**

In this question, we will study and compare two widely used dimensionality reduction methods in single-cell RNA-seq analysis: (i) Principal Components Analysis (PCA) and (ii) t-distributed Stochastic Neighbor Embedding (t-SNE). The goal is to understand how the objective functions of these methods determine whether they preserve global or local structure in high-dimensional data.

Throughout this problem, assume that the data matrix $X \in \mathbb{R}^{n \times d}$ has been mean-centered (i.e., each feature has zero mean).

Your task

Assume that the data matrix $X \in \mathbb{R}^{n \times d}$ is **mean-centered**.

(a) (5 points) **PCA: Variance maximization formulation.**
Consider projecting each data point $x_i \in \mathbb{R}^d$ onto a one-dimensional direction $w \in \mathbb{R}^d$ with $\|w\|_2 = 1$, producing

$$z_i = w^\top x_i.$$

Show that the variance of the projected data can be written as

$$\mathrm{Var}(z) = w^\top \Sigma w,$$

where $\Sigma = \frac{1}{n} X^\top X$ is the sample covariance matrix.

Solution

(b) (5 points) **PCA: Eigenvalue problem.**
PCA seeks the direction w that maximizes $w^\top \Sigma w$ subject to $\|w\|_2 = 1$.

i. Using a Lagrange multiplier, show that the optimal solution satisfies

$$\Sigma w = \lambda w.$$

ii. Explain why the eigenvector corresponding to the largest eigenvalue is selected as the first principal component.

Solution

(c) (5 points) **PCA and global structure.**
PCA can also be viewed as minimizing the total squared reconstruction error

$$\sum_{i=1}^{n} \|x_i - \hat{x}_i\|_2^2,$$

where $\hat{x}_i$ denotes the projection of x_i onto the low-dimensional subspace.
Explain why this objective makes PCA particularly sensitive to *global* structure in the data. In your answer, discuss how distances between far-apart points influence the solution.

Solution

(d) (8 points) **t-SNE: High-dimensional similarities**

In t-SNE, similarities between points in the original space are defined as conditional probabilities:

$$p_{j|i} = \frac{\exp\left(-\|x_i - x_j\|^2/(2\sigma_i^2)\right)}{\sum_{k\neq i}\exp\left(-\|x_i - x_k\|^2/(2\sigma_i^2)\right)}.$$

Explain the intuition behind this definition. What does it mean for $p_{j|i}$ to be large or small?

Solution

(e) (12 points) **t-SNE: KL divergence, asymmetry, and locality**

In t-SNE, similarities in the low-dimensional embedding are defined as

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k\neq l}(1 + \|y_k - y_l\|^2)^{-1}}.$$

The low-dimensional embedding is obtained by minimizing the Kullback–Leibler (KL) divergence between the high-dimensional and low-dimensional similarity distributions:

$$\mathcal{L} = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}}.$$

i. Explain how the weighting by p_{ij} affects which pairs of points contribute most to the loss.
ii. Consider a pair of points that are far apart in the high-dimensional space but close together in the low-dimensional embedding. Explain why this typically does *not* incur a large penalty.
iii. Qualitatively describe what would change if t-SNE minimized $\mathrm{KL}(Q\|P)$ instead of $\mathrm{KL}(P\|Q)$.

Solution

(f) (5 points) **Comparison: PCA vs. t-SNE**

Why is PCA generally considered a *global* dimensionality reduction method, while t-SNE is considered a *local* dimensionality reduction method?

Your answer should explicitly reference the objective functions of both methods.

Solution

2. **[40 points] Clustering**

In this question, we will perform and compare two clustering algorithms on real single-cell RNA-seq data: (i) K-means clustering and (ii) graph-based Leiden clustering. The goal is to understand how different clustering assumptions affect both quantitative performance and biological interpretability in single-cell analysis.

Data Description

We provide a single-cell RNA-seq dataset in an AnnData object (file: `sc_pca.h5ad`). The data are derived from:

> Smillie, Christopher S., et al. *"Intra- and inter-cellular rewiring of the human colon during ulcerative colitis."* Cell 178.3 (2019): 714–730.

The AnnData object contains:

- **Cells:** 34,162 cells.
- `obsm` (embeddings): `X_pca`, a 30-dimensional PCA embedding for each cell.
- `obs` (cell metadata): a column `merged_annotation` with 9 cell type labels: Plasma, Follicular, CD4+ Memory, CD8+ LP, Enterocytes, TA, Macrophages, Immature Goblet, Immature Enterocytes 1.

All clustering should be performed using the PCA embeddings (`X_pca`) only. The cell type annotations are provided *for evaluation and interpretation purposes only.*

Notes:

- You may use *any* programming language and packages (e.g., Scanpy, Seurat, scikit-learn) that you find useful.
- No skeleton code is provided. Your code will not be graded by an autograder; instead, you should submit your code together with your written solutions. Partial credit may be awarded for some correct implementations even if final results are incorrect.
- For K-means clustering, you should use K-means++ initialization or multiple random restarts and select the best solution.

Your task

(a) (15 points) **K-means clustering.**

K-means clustering partitions the data into k clusters by minimizing the within-cluster sum of squared errors (SSE) in Euclidean space. Given cluster assignments $\{C_1, \dots, C_k\}$ and cluster centroids $\{\mu_1, \dots, \mu_k\}$, the SSE is defined as:

$$\text{SSE} = \sum_{c=1}^{k} \sum_{x_i \in C_c} \|x_i - \mu_c\|^2.$$

i. (5 points) Run K-means clustering for $k = 1, 2, \dots, 15$ on `X_pca`. Plot SSE versus k and identify the elbow point. Briefly explain what the elbow point represents and why it may be ambiguous for single-cell data.

ii. (5 points) Run K-means with $k = 10$. To ensure stable results, use either K-means++ initialization or run K-means multiple times with different random initializations and select the solution with the lowest SSE. Report the final SSE value.

iii. (5 points) Compute the Adjusted Rand Index (ARI) between the K-means cluster assignments ($k = 10$) and the ground-truth labels `merged_annotation`. The ARI between two partitions U and V is defined as:

$$\mathrm{ARI} = \frac{\mathrm{RI} - \mathbb{E}[\mathrm{RI}]}{\max(\mathrm{RI}) - \mathbb{E}[\mathrm{RI}]},$$

where RI is the Rand Index measuring pairwise agreement between clusterings. Report and interpret the ARI value.

(b) (15 points) **Graph-based Leiden clustering.**

Graph-based clustering methods operate on a neighborhood graph and can capture clusters with arbitrary shapes. In this part, you will apply Leiden clustering.

i. (5 points) Construct a k-nearest neighbor (kNN) graph from `X_pca` using $k = 20$ neighbors per cell. Clearly state the distance metric used.

ii. (5 points) Run the Leiden algorithm on the kNN graph with resolution parameter `resolution = 0.1`. Report the number of clusters obtained.

iii. (5 points) Compute the ARI between the Leiden cluster assignments (resolution = 0.1) and `merged_annotation`. Report and interpret the ARI value.

Solution

(c) (10 points) **Visualization and comparison.**

UMAP is commonly used to visualize single-cell data and clustering results.

i. (5 points) Plot a UMAP embedding colored by K-means clusters using $k = 4$.

ii. (5 points) Plot the same UMAP embedding colored by Leiden clusters using `resolution = 0.01`.

Question: Based on the UMAP visualizations of K-means ($k = 4$) and Leiden (resolution = 0.01), what is a potential disadvantage of K-means–based clustering methods in single-cell RNA-seq analysis? Briefly explain your reasoning.

Solution

3. **[60 points] Motif Finding**

In lecture we were introduced to an algorithm used for finding motifs in DNA sequences based on Expectation Maximization (EM). This algorithm forms the basis of the MEME Suite (**M**ultiple **E**M for **M**otif **E**lucidation), one of the most widely used softwares in genomics. Several good papers are available for understanding the algorithm, including the ones here and here.

Consider a biological motif of length W, $M = (M_1, \ldots, M_W)$, where $M_i \in \{A, C, G, T\}$. Our model for biological motifs is that each M_i is a Multinoulli-distributed random with its own probability distribution over the nucleotides $A, C, G,$ and T. We can equivalently represent this motif as a position weight matrix $(M)_{ij}$, for which

$$M_{ij} = \mathbb{P}(M_j = i),$$

where $j = 1, \ldots, W$ and $i \in \{A, C, G, T\}$. Consider the PWM below for a motif of length $W = 6$:

$$M = \begin{array}{c} \\ A \\ C \\ G \\ T \end{array} \begin{array}{c} \begin{array}{cccccc} M_1 & M_2 & M_3 & M_4 & M_5 & M_6 \end{array} \\ \begin{bmatrix} 0.8 & 0.1 & 0 & 0.9 & 0 & 0.3 \\ 0 & 0.4 & 0.05 & 0.03 & 0.1 & 0.2 \\ 0.2 & 0 & 0.95 & 0.02 & 0.1 & 0.1 \\ 0 & 0.5 & 0 & 0.05 & 0.8 & 0.4 \end{bmatrix} \end{array} \tag{1}$$

One of the key assumptions we make when modeling motifs is that $M_i \perp M_j$ for $i \neq j$; that is, the distributions of the nucleotides in each position of the motif are independent of one another.

(a) Find the probability $\mathbb{P}(M = ACCTTA)$.

Solution

(b) Provide the Consensus Sequence corresponding to the PWM above; i.e., the sequence $m = (m_1, \ldots, m_5)$ such that $\mathbb{P}(M = m)$ is maximized.

Solution

(c) Which of the following sequence logos accurately represents the PWM? Briefly describe why the other logos are incorrect.

1) 

2)

3)

Solution

(d) The height of the sequence logo is often scaled by the information content at a given position i. The information content is given by $R_i = log_2(4) - (H_i + e_n)$, where H_i is the entropy of the position, and $H_i = -\Sigma_j [M_{i,j} * log_2 M_{i,j}]$.
Which position in the PWM has the greatest entropy? How is entropy related to conservation?

Solution

(e) One of the advantages of the MEME suite is that, for a given DNA sequence, it can detect the most likely positions to find the motif learned in the PWM. To do so, it converts the PWM into a matrix of log likehood ratios using the formula $LLR(s) = log_2[\frac{Pr(s|M)}{Pr(s|M_{bg})})]$. The log likehood score that a given position i is a start position of the motif is equivalent to $S(X) = \sum_{i=x_i}^{i=x_i+w} log_2[\frac{M_{ij}}{M_{ij}^{bg}}]$.

One method of determining if a position is included in the motif is to use a decision rule. For example, we could use : 'X is a true instance of M if $LLR(X) > 0$, or equivalently, if $\mathcal{S}(X) > 0$'. Is this a statistically sound approach? Why or why not?

Solution

(f) An alternative approach to determine where or not a position i contains the motif is to use hypothesis testing to calculate the p-value for S(p). We can define the hypotheses as:

- H_0: X is drawn from the background distribution M^{bg}
- H_1: X is drawn from the motif distribution M

Briefly describe a method for calculating the test statistic. Hint: one definition of p-value is the probability of receiving a value as or even more extreme than S(X) under the null hypothesis.

Solution